



南方科技大学  
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 高等数学(下) 开课单位: 数学系

考试时长: 120 分钟 命题教师: \_\_\_\_\_

| 题号 | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    |
|----|------|------|------|------|------|------|------|------|
| 分值 | 20 分 | 20 分 | 10 分 | 10 分 | 10 分 | 10 分 | 10 分 | 10 分 |

本试卷共 8 道大题, 满分 100 分. (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

注意: 本试卷里的中文为直译 (即完全按英文字面意思直接翻译), 所有数学词汇的定义请参照教材 (Thomas' Calculus, 13th Edition) 中的定义. 如果其中有些数学词汇的定义不同于中文书籍 (比方说同济大学的高等数学教材) 里的定义, 以教材 (Thomas' Calculus, 13th Edition) 中的定义为准.

1. (20pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

(1) If  $P_1$  and  $P_2$  are the following planes:

$$P_1 := \{(x, y, z) : x + 2y - z = \sqrt{2}/2\},$$

$$P_2 := \{(x, y, z) : x + y = -\sqrt{2}/2\},$$

then the angle  $\theta$  between  $P_1$  and  $P_2$  is:

(A)  $\frac{\pi}{2}$ . (B)  $\frac{\pi}{3}$ .

(C)  $\frac{\pi}{4}$ . (D)  $\frac{\pi}{6}$ .

(2) The arc length of the curve  $\mathbf{r}(t) = \frac{2}{3}t^{\frac{3}{2}}\mathbf{i} + \frac{2}{3}(2-t)^{\frac{3}{2}}\mathbf{j} + (t-1)\mathbf{k}$  for  $\frac{1}{4} \leq t \leq \frac{1}{2}$  is

(A)  $\frac{1}{4}$ . (B)  $\frac{\sqrt{2}}{4}$ .

(C)  $\frac{\sqrt{3}}{4}$ . (D)  $\frac{1}{2}$ .

(3) Let  $f(x, y) = \begin{cases} \frac{\ln(1+xy)}{x}, & x \neq 0 \\ a, & x = 0 \end{cases}$ , then

(A) If  $a = 1$ , then  $f(x, y)$  is continuous at  $(0, 2)$ .

(B) If  $a = 2$ , then  $f(x, y)$  is discontinuous at  $(0, 2)$ .

(C) If  $a = 1$ , then  $f(x, y)$  is continuous at  $(0, 1)$ .

(D) If  $a = 2$ , then  $f(x, y)$  is continuous at  $(0, 1)$ .

(4) If  $0 \leq a_n \leq \frac{1}{n}$  ( $n = 1, 2, \dots$ ), which of the following series must be convergent ?

(A)  $\sum_{n=1}^{\infty} (-1)^n \frac{a_n}{(1 + \ln n)^2}$ . (B)  $\sum_{n=1}^{\infty} (-1)^n a_n$ .

(C)  $\sum_{n=1}^{\infty} a_n$  .

(D)  $\sum_{n=1}^{\infty} (-1)^n \frac{a_n}{1 + \ln n}$  .

(5) If  $\sum_{n=1}^{\infty} a_n x^n$  converges at  $x = -2$ , then

(A)  $\sum_{n=1}^{\infty} a_n 2^n$  converges.

(B)  $\sum_{n=1}^{\infty} a_n 2^n$  diverges.

(C)  $\sum_{n=1}^{\infty} n a_n$  converges.

(D)  $\sum_{n=1}^{\infty} n a_n$  diverges.

2. (20 pts) Fill in the blanks.

(1) Let the polar equation be  $r = \csc \theta e^{r \cos \theta}$ . Its equivalent Cartesian equations is \_\_\_\_\_.

(2) The distance from the point  $S(1, 1, 1)$  to the plane  $x + 2y + 3z + 4 = 0$  is \_\_\_\_\_.

(3) A particle is traveling with acceleration  $\mathbf{a}(t) = \langle e^t, t, \sin 2t \rangle$ . At  $t = 0$ , it was at the origin and its initial velocity is  $\mathbf{v}(0) = \langle 1, 0, 0 \rangle$ . The position function of the particle is \_\_\_\_\_.

(4)  $\lim_{x \rightarrow 0} \frac{3 \sin(2x) - 2 \sin(3x)}{6x^3 + x^4} = \underline{\hspace{2cm}}.$

(5) The series

$$S = -\ln 2 + \frac{\ln^2 2}{2!} + \cdots + \frac{(-1)^n \ln^n 2}{n!} + \cdots$$

converges and has the sum \_\_\_\_\_.

3. (10 pts)

(1) Find the interval of convergence for the series  $\sum_{n=1}^{\infty} \frac{\ln n}{n} (x-1)^n$ .

(2) For what value of  $x$  the above series converges absolutely, or conditionally ?

4. (10 pts) Find the limit if it exists. If it does not exist, explain why.

(1)  $\lim_{(x,y) \rightarrow (0,0)} \frac{y^2 \sin^2 x}{x^4 + y^4}.$

(2)  $\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{\sqrt{x^2 + 2y^2}}.$

5. (10 pts) Let  $L_1: x = t, y = 1, z = 1 + t$  and  $L_2: x = 1 + 2s, y = 0, z = 2s$ . Is there one plane in which both lines lie ? If so, find the equation of the plane. If not, give your reason.

6. (10 pts) Assume that  $a$  and  $b$  are real numbers, and  $0 < b < a < 1$ .

(a) Determine whether the series

$$\begin{aligned} & \frac{a}{\sqrt{(1+a)(1+b)}} + \frac{b^2}{\sqrt{(1+a^2)(1+b^2)}} + \frac{a^3}{\sqrt{(1+a^3)(1+b^3)}} \\ & + \frac{b^4}{\sqrt{(1+a^4)(1+b^4)}} + \cdots + \frac{c_n}{\sqrt{(1+a^n)(1+b^n)}} + \cdots, \end{aligned}$$

where

$$c_n = \begin{cases} a^n, & \text{if } n \text{ is odd,} \\ b^n, & \text{if } n \text{ is even,} \end{cases}$$

converges or diverges. **Justify your answer!**

(b) Find the limit:

$$\lim_{n \rightarrow \infty} \left( \frac{a^n}{n^2} + \frac{b^n}{n} \right)^{1/n}.$$

7. (10 pts) Let  $C$  be the curve given by

$$\mathbf{r}(t) = \sqrt{2} \cos t \mathbf{i} + \sin t \mathbf{j} + \sin t \mathbf{k}.$$

(a) Find the unit tangent vector  $\mathbf{T}(t)$  and principal unit normal vector  $\mathbf{N}(t)$  for  $C$ .

(b) Find the curvature  $\kappa(t)$ .

8. (10 pts) Using known Taylor series expansions, write the Taylor series for the function

$$f(x) = \frac{4}{x^2 - 2x + 5} + \ln x$$

at  $x = 1$  in the interval  $(0, 2)$ .

一、 (20分) 单项选择题:

(1) 定义平面  $P_1$  和  $P_2$  如下:

$$P_1 := \{(x, y, z): x + 2y - z = \sqrt{2}/2\},$$

$$P_2 := \{(x, y, z): x + y = -\sqrt{2}/2\},$$

则平面  $P_1$  和  $P_2$  的夹角  $\theta$  为:

- (A)  $\frac{\pi}{2}$ . (B)  $\frac{\pi}{3}$ .  
(C)  $\frac{\pi}{4}$ . (D)  $\frac{\pi}{6}$ .

(2) 曲线  $\mathbf{r}(t) = \frac{2}{3}t^{\frac{3}{2}}\mathbf{i} + \frac{2}{3}(2-t)^{\frac{3}{2}}\mathbf{j} + (t-1)\mathbf{k}$ ,  $\frac{1}{4} \leq t \leq \frac{1}{2}$  的弧长为

- (A)  $\frac{1}{4}$ . (B)  $\frac{\sqrt{2}}{4}$ .  
(C)  $\frac{\sqrt{3}}{4}$ . (D)  $\frac{1}{2}$ .

(3) 设  $f(x, y) = \begin{cases} \frac{\ln(1+xy)}{x}, & x \neq 0 \\ a, & x = 0 \end{cases}$ , 则

- (A) 若  $a = 1$ , 则  $f(x, y)$  在点  $(0, 2)$  处连续.  
(B) 若  $a = 2$ , 则  $f(x, y)$  在点  $(0, 2)$  处不连续.  
(C) 若  $a = 1$ , 则  $f(x, y)$  在点  $(0, 1)$  处连续.  
(D) 若  $a = 2$ , 则  $f(x, y)$  在点  $(0, 1)$  处连续.

(4) 若  $0 \leq a_n \leq \frac{1}{n}$  ( $n = 1, 2, \dots$ ), 则下列级数中肯定收敛的是

- (A)  $\sum_{n=1}^{\infty} (-1)^n \frac{a_n}{(1 + \ln n)^2}$ . (B)  $\sum_{n=1}^{\infty} (-1)^n a_n$ .  
(C)  $\sum_{n=1}^{\infty} a_n$ . (D)  $\sum_{n=1}^{\infty} (-1)^n \frac{a_n}{1 + \ln n}$ .

(5) 若级数  $\sum_{n=1}^{\infty} a_n x^n$  在  $x = -2$  处收敛, 则

- (A)  $\sum_{n=1}^{\infty} a_n 2^n$  收敛. (B)  $\sum_{n=1}^{\infty} a_n 2^n$  发散.  
(C)  $\sum_{n=1}^{\infty} n a_n$  收敛. (D)  $\sum_{n=1}^{\infty} n a_n$  发散.

二、 (20分) 填空题:

(1) 设曲线的极坐标方程是  $r = \csc \theta e^{r \cos \theta}$ , 则曲线的平面直角坐标系方程为 \_\_\_\_\_.

(2) 从点  $S(1, 1, 1)$  到平面  $x + 2y + 3z + 4 = 0$  的距离是 \_\_\_\_\_.

(3) 一质点的加速度为  $\mathbf{a}(t) = \langle e^t, t, \sin 2t \rangle$ , 在  $t = 0$  时, 它从原点出发, 初始速度为  $\mathbf{v}(0) = \langle 1, 0, 0 \rangle$ . 则此质点的位置函数为 \_\_\_\_\_.

(4)  $\lim_{x \rightarrow 0} \frac{3 \sin(2x) - 2 \sin(3x)}{6x^3 + x^4} =$  \_\_\_\_\_.

(5) 级数

$$S = -\ln 2 + \frac{\ln^2 2}{2!} + \cdots + \frac{(-1)^n \ln^n 2}{n!} + \cdots$$

收敛, 且该级数的和为 \_\_\_\_\_.

三、 (10分)

(1) 求级数  $\sum_{n=1}^{\infty} \frac{\ln n}{n} (x-1)^n$  的收敛域.

(2) 上述级数在哪些点上绝对收敛? 在哪些点上条件收敛?

四、 (10分) 对于下列极限, 若极限存在, 求其值; 若极限不存在, 说明理由.

(1)  $\lim_{(x,y) \rightarrow (0,0)} \frac{y^2 \sin^2 x}{x^4 + y^4}.$

(2)  $\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{\sqrt{x^2 + 2y^2}}.$

五、 (10分) 设两条直线定义为  $L_1: x = t, y = 1, z = 1 + t$  和  $L_2: x = 1 + 2s, y = 0, z = 2s$ . 这两条直线是否共面? 若是, 请写出该平面的方程; 若不是, 给出理由.

六、 (10分) 若  $a$  和  $b$  为实数且满足  $0 < b < a < 1$ .

(a) 判断

$$\begin{aligned} & \frac{a}{\sqrt{(1+a)(1+b)}} + \frac{b^2}{\sqrt{(1+a^2)(1+b^2)}} + \frac{a^3}{\sqrt{(1+a^3)(1+b^3)}} \\ & + \frac{b^4}{\sqrt{(1+a^4)(1+b^4)}} + \cdots + \frac{c_n}{\sqrt{(1+a^n)(1+b^n)}} + \cdots, \end{aligned}$$

这里

$$c_n = \begin{cases} a^n, & n \text{ 是奇数} \\ b^n, & n \text{ 是偶数} \end{cases}$$

是否收敛, 并给出理由.

(b) 求极限

$$\lim_{n \rightarrow \infty} \left( \frac{a^n}{n^2} + \frac{b^n}{n} \right)^{1/n}.$$

七、 (10分) 已知曲线  $C$  的参数方程是

$$\mathbf{r}(t) = \sqrt{2} \cos t \mathbf{i} + \sin t \mathbf{j} + \sin t \mathbf{k}.$$

(a) 求此曲线  $C$  的单位切向量  $\mathbf{T}(t)$  和单位主法向量  $\mathbf{N}(t)$ .

(b) 求曲率  $\kappa(t)$ .

八、 (10分) 使用已知的泰勒级数, 写出函数

$$f(x) = \frac{4}{x^2 - 2x + 5} + \ln x$$

在  $x = 1$  处的泰勒级数, 这里  $x \in (0, 2)$ .

1. (20pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

D (1) If  $P_1$  and  $P_2$  are the following planes:

$$P_1 := \{(x, y, z) : x + 2y - z = \sqrt{2}/2\},$$

$$P_2 := \{(x, y, z) : x + y = -\sqrt{2}/2\},$$

then the angle  $\theta$  between  $P_1$  and  $P_2$  is:

(A)  $\frac{\pi}{2}$ .

(B)  $\frac{\pi}{3}$ .

(C)  $\frac{\pi}{4}$ .

(D)  $\frac{\pi}{6}$ .

$$P_1 : \vec{n}_1 = (1, 2, -1) \quad \cos \theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1| |\vec{n}_2|} = \frac{3}{\sqrt{6} \sqrt{2}} = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$$

$$P_2 : \vec{n}_2 = (1, 1, 0)$$

C (2) The arc length of the curve  $\mathbf{r}(t) = \frac{2}{3}t^{\frac{3}{2}}\mathbf{i} + \frac{2}{3}(2-t)^{\frac{3}{2}}\mathbf{j} + (t-1)\mathbf{k}$  for  $\frac{1}{4} \leq t \leq \frac{1}{2}$  is

(A)  $\frac{1}{4}$ .

(B)  $\frac{\sqrt{2}}{4}$ .

(C)  $\frac{\sqrt{3}}{4}$ .

(D)  $\frac{1}{2}$ .

$$L = \int_{\frac{1}{4}}^{\frac{1}{2}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \int_{\frac{1}{4}}^{\frac{1}{2}} \sqrt{t + (2-t) + 1} dt = \frac{\sqrt{3}}{4}$$

C (3) Let  $f(x, y) = \begin{cases} \frac{\ln(1+xy)}{x}, & x \neq 0 \\ a, & x = 0 \end{cases}$ , then

(A) If  $a = 1$ , then  $f(x, y)$  is continuous at  $(0, 2)$ .

(B) If  $a = 2$ , then  $f(x, y)$  is discontinuous at  $(0, 2)$ .

(C) If  $a = 1$ , then  $f(x, y)$  is continuous at  $(0, 1)$ .

(D) If  $a = 2$ , then  $f(x, y)$  is continuous at  $(0, 1)$ .

$$C : y=kx \quad \lim_{(x,y) \rightarrow (0,1)} \frac{\ln(1+xy)}{x} = \lim_{(x,y) \rightarrow (0,1)} \frac{xy}{x} = 1 = a = f(0,1)$$

A (4) If  $0 \leq a_n \leq \frac{1}{n}$  ( $n = 1, 2, \dots$ ), which of the following series must be convergent?

(A)  $\sum_{n=1}^{\infty} (-1)^n \frac{a_n}{(1 + \ln n)^2}$ .

(B)  $\sum_{n=1}^{\infty} (-1)^n a_n$ .

(C)  $\sum_{n=1}^{\infty} a_n$ .

(D)  $\sum_{n=1}^{\infty} (-1)^n \frac{a_n}{1 + \ln n}$ .

$$(A) \quad \frac{a_n}{(1 + \ln n)^2} \leq \frac{1}{n(1 + \ln n)^2} \quad \int_1^{\infty} \frac{1}{x(1 + \ln x)^2} dx = \int_1^{\infty} \frac{1}{(1 + \ln x)^2} d \ln x$$

$$(C) \quad a_n = \frac{1}{n}$$

$$(B) \quad \begin{cases} a_{2k+1} = \frac{1}{2k+1} \\ a_{2k} = \frac{1}{4k^2} \end{cases}$$

$$= -\frac{1}{1 + \ln x} \Big|_1^{+\infty}$$

$$= 1 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n(1 + \ln n)^2} < \infty$$

$$(D) \quad \begin{cases} a_{2k+1} = \frac{1}{2k+1} \\ a_{2k} = \frac{1}{4k^2} \end{cases} \text{ diverges} \quad \int_1^{\infty} \frac{1}{x(1 + \ln x)} dx = +\infty$$

C (5) If  $\sum_{n=1}^{\infty} a_n x^n$  converges at  $x = -2$ , then

(A)  $\sum_{n=1}^{\infty} a_n 2^n$  converges.

(B)  $\sum_{n=1}^{\infty} a_n 2^n$  diverges.

(C)  $\sum_{n=1}^{\infty} n a_n$  converges.

(D)  $\sum_{n=1}^{\infty} n a_n$  diverges.

$$|x| < 2 \quad \sum_{n=1}^{\infty} a_n x^n \text{ converges}$$

Term-by-Term Differentiation:

$$\sum_{n=1}^{\infty} n a_n x^{n-1} \quad |x| < 2$$

$$x=1 \quad \sum_{n=1}^{\infty} n a_n \text{ converges}$$

2. (20 pts) Fill in the blanks.

(1) Let the polar equation be  $r = \csc \theta e^{r \cos \theta}$ . Its equivalent Cartesian equations is  $y = e^x$ .

$$r = \csc \theta e^{r \cos \theta}$$

$$r \sin \theta = e^{r \cos \theta}$$

$$y = e^x$$

(2) The distance from the point  $S(1, 1, 1)$  to the plane  $x + 2y + 3z + 4 = 0$  is  $\frac{5\sqrt{14}}{7}$ .

$$d = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|1 + 2 + 3 + 4|}{\sqrt{1^2 + 2^2 + 3^2}} = \frac{10}{\sqrt{14}} = \frac{5\sqrt{14}}{7} \quad d = \left| \vec{PS} \cdot \frac{\vec{n}}{|\vec{n}|} \right|$$

(3) A particle is traveling with acceleration  $\mathbf{a}(t) = \langle e^t, t, \sin 2t \rangle$ . At  $t = 0$ , it was at the origin and its initial velocity is  $\mathbf{v}(0) = \langle 1, 0, 0 \rangle$ . The position function of the particle is  $(e^t - 1, \frac{1}{2}t^2, -\frac{1}{4}\sin 2t + \frac{1}{2}t)$

$$\mathbf{v}(t) = \langle e^t + C_1, \frac{1}{2}t^2 + C_2, -\frac{1}{2}\cos 2t + C_3 \rangle \quad \mathbf{v}(0) = \langle 1, 0, 0 \rangle$$

$$\mathbf{v}(t) = \langle e^t, \frac{1}{2}t^2, -\frac{1}{2}\cos 2t + \frac{1}{2} \rangle$$

$$\mathbf{r}(t) = \langle e^t + C_1', \frac{1}{6}t^3 + C_2', -\frac{1}{4}\sin 2t + \frac{1}{2}t + C_3' \rangle$$

$$\mathbf{r}(0) = \langle 0, 0, 0 \rangle \quad \mathbf{r}(t) = \langle e^t - 1, \frac{1}{6}t^3, -\frac{1}{4}\sin 2t + \frac{1}{2}t \rangle$$

(4)  $\lim_{x \rightarrow 0} \frac{3 \sin(2x) - 2 \sin(3x)}{6x^3 + x^4} = \underline{\frac{5}{6}}$ .

$$= \lim_{x \rightarrow 0} \frac{3(2x - \frac{(2x)^3}{3!} + o(x^3)) - 2(3x - \frac{(3x)^3}{3!} + o(x^3))}{6x^3 + x^4}$$

$$= \lim_{x \rightarrow 0} \frac{6x - \frac{8x^3}{2} - 6x + 9x^3 + o(x^3)}{6x^3 + x^4} = \lim_{x \rightarrow 0} \frac{5x^3 + o(x^3)}{6x^3 + x^4} = \frac{5}{6}$$



(5) The series

$$S = -\ln 2 + \frac{\ln^2 2}{2!} + \dots + \frac{(-1)^n \ln^n 2}{n!} + \dots$$

converges and has the sum  $e^{-\ln 2} - 1$

$$S = \sum_{n=1}^{\infty} \frac{(-1)^n (\ln 2)^n}{n!} = \sum_{n=1}^{\infty} \frac{(-\ln 2)^n}{n!} = e^{-\ln 2} - 1$$

3. (10 pts)

(1) Find the interval of convergence for the series  $\sum_{n=1}^{\infty} \frac{\ln n}{n} (x-1)^n$ .

(2) For what value of  $x$  the above series converges absolutely, or conditionally?

$$(1) \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{\ln n}{n} (x-1)^n \right|} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{\ln n}}{\sqrt[n]{n}} |x-1| = |x-1|$$

$$|x-1| < 1 \Rightarrow 0 < x < 2 \quad \sum_{n=1}^{\infty} \frac{\ln n}{n} (x-1)^n \text{ converges absolutely}$$

$$x-1 = 1 \quad \sum_{n=1}^{\infty} \frac{\ln n}{n}$$

$$\frac{\ln n}{n} > \frac{1}{n} \quad \text{when } n \geq 3$$

$$\sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges} \Rightarrow \sum_{n=1}^{\infty} \frac{\ln n}{n} \text{ diverges}$$

$$x-1 = -1 \quad \sum_{n=1}^{\infty} (-1)^n \frac{\ln n}{n}$$

$$f(x) = \frac{\ln x}{x} \quad f'(x) = \frac{1-\ln x}{x^2} \quad x > e \quad f(x) \text{ non-increasing}$$

$$n \geq 3 \quad \frac{\ln(n+1)}{n+1} \leq \frac{\ln n}{n}$$

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0 \quad \frac{\ln n}{n} > 0 \quad n \geq 2$$

By Alternative Series Test,

$$\sum_{n=1}^{\infty} \frac{\ln n}{n} (-1)^n \text{ converges}$$

Interval of convergence is  $[0, 2)$

$$(2) \quad x \in (0, 2) \quad \sum_{n=1}^{\infty} \frac{\ln n}{n} (x-1)^n \text{ converges absolutely}$$

$$x=0 \quad \sum_{n=1}^{\infty} \frac{\ln n}{n} (x-1)^n \text{ converges conditionally.}$$

5. (10 pts) Let  $L_1: x=t, y=1, z=1+t$  and  $L_2: x=1+2s, y=0, z=2s$ . Is there one plane in which both lines lie? If so, find the equation of the plane. If not, give your reason.

$$\text{Pf. Direction of } L_1 : \vec{d}_1 = (1, 0, 1)$$

$$\text{Direction of } L_2 : \vec{d}_2 = (2, 0, 2)$$

$$\vec{d}_2 = 2\vec{d}_1$$

$L_1$  and  $L_2$  are parallel.

They are in the same plane.

Take a point  $P_1(0, 1, 1)$  in  $L_1$ , another point  $P_2(1, 0, 0)$  in  $L_2$ .

We have  $\vec{P_1P_2} = (1, -1, -1)$

Then we calculate the normal of the plane.

$$\vec{n} = \vec{d_1} \times \vec{P_1P_2} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{vmatrix} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

Equation of the plane

$$x + 2y - z = D$$

Substitute  $P_1$  into the equation,

$$D = 1$$

The equation of plane is  $x + 2y - z = 1$ .

6. (10 pts) Assume that  $a$  and  $b$  are real numbers, and  $0 < b < a < 1$ .

(a) Determine whether the series

$$\frac{a}{\sqrt{(1+a)(1+b)}} + \frac{b^2}{\sqrt{(1+a^2)(1+b^2)}} + \frac{a^3}{\sqrt{(1+a^3)(1+b^3)}} \\ + \frac{b^4}{\sqrt{(1+a^4)(1+b^4)}} + \dots + \frac{c_n}{\sqrt{(1+a^n)(1+b^n)}} + \dots,$$

where

$$c_n = \begin{cases} a^n, & \text{if } n \text{ is odd,} \\ b^n, & \text{if } n \text{ is even,} \end{cases}$$

converges or diverges. Justify your answer!

Converges

$n$  is odd Let  $n = 2k+1$

$$\frac{c_{2k+1}}{\sqrt{(1+a^{2k+1})(1+b^{2k+1})}} = \frac{a^{2k+1}}{\sqrt{(1+a^{2k+1})(1+b^{2k+1})}} \leq a^{2k+1}$$

$n$  is even Let  $n = 2k$

$$\frac{c_{2k}}{\sqrt{(1+a^{2k})(1+b^{2k})}} = \frac{b^{2k}}{\sqrt{(1+a^{2k})(1+b^{2k})}} \leq b^{2k} \leq a^{2k}$$

$$\Rightarrow \frac{c_n}{\sqrt{(1+a^n)(1+b^n)}} \leq a^n \quad \sum_{n=1}^{\infty} a^n = \frac{a}{1-a} < \infty$$

$\Rightarrow$  By Comparison Test,  $\sum_{n=1}^{\infty} \frac{C_n}{\sqrt{(1+a^n)(1+b^n)}}$  converges.

(b) Find the limit:

$$\lim_{n \rightarrow \infty} \left( \frac{a^n}{n^2} + \frac{b^n}{n} \right)^{1/n}.$$

Pf.  $\frac{a^n}{n^2} \leq \frac{a^n}{n^2} + \frac{b^n}{n} \leq \frac{a^n}{n} + \frac{b^n}{n} \leq \frac{2a^n}{n}$

So  $\left( \frac{a^n}{n^2} \right)^{\frac{1}{n}} \leq \left( \frac{a^n}{n^2} + \frac{b^n}{n} \right)^{\frac{1}{n}} \leq \left( \frac{2a^n}{n} \right)^{\frac{1}{n}}$

$$\lim_{n \rightarrow \infty} \left( \frac{a^n}{n^2} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left( \frac{2a^n}{n} \right)^{\frac{1}{n}} = a$$

By Sandwich Theorem

$$\lim_{n \rightarrow \infty} \left( \frac{a^n}{n^2} + \frac{b^n}{n} \right)^{\frac{1}{n}} = a.$$